

Causation vs. Correlation

What Makes A Relationship “Causal”?

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POLS 222: Slide Deck #5

Where We Left Off

Theories provide simpler explanations to complex problems.

- They are the concise, general answers to the research question.

We can generate theories using induction (from observations to theories).

We can test theories using deduction (from theory to observation).

Good theories must be: testable, falsifiable, logically complete, logically consistent, and empirically supported.

Parts of a theory:

1. The explanation (the short answer to the research question)
2. The mechanism (X causes Y, but *how* exactly?)
3. The assumption (can be implicit or explicit, must be reasonable)
4. The conditions (what is the scope over which your answer is true?)

One Hypothesis, Two Hypotheses

A **hypothesis** is a specific, testable expectation about reality, based on the argument.

- It's a statement of the relationship that you expect to find between an IV and your DV.

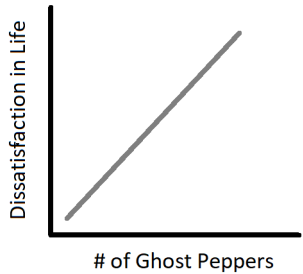
How hypotheses can look:

- The higher (lower) the X, the higher (lower) the Y.
- As X increases (or decreases), Y increases (or decreases).
- When X is true (or false), Y is true (or false)
- As X increases (or decreases), the probability of Y being true increases (or decreases).

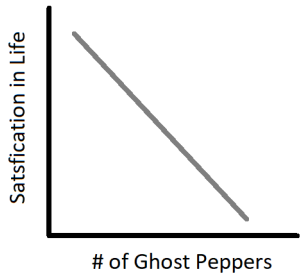
Let's start with a statement: "The higher the X, the more likely it is that Y will occur because more X makes W happen, which incentivizes actors to do Y."

- Everything before "**because**" is the hypothesis, and everything after is the causal theory.

Positive vs. Negative Relationships

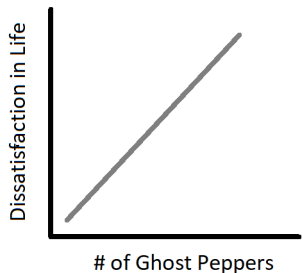


Positive
(Direct)

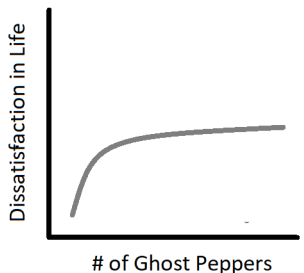


Negative
(Indirect/inverse)

Linear vs. Nonlinear Relationships

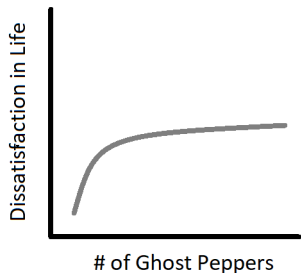


Linear

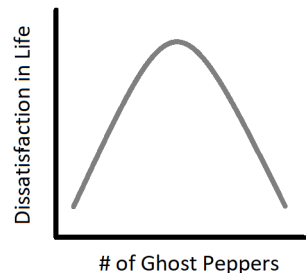


Nonlinear

Monotonic vs. Nonmonotonic Relationships



Monotonic



Non-monotonic

But Are These Causal Relationships?

Correlation does not imply causation.

Nomothetic causation:

- The causal link is probabilistic, not deterministic.
- An outlier or two doesn't "disprove" or "refute" the causal theory.
- ...a butt load of outliers might, though...
- To say that X_1 causes Y does not exclude other things (e.g. X_2) from being causally linked, too.
 - but the relationship between X_1 and X_2 matters here, because we have to look out for confounders and interveners

Rules for Establishing Causality

1. Variation in the causal variable
2. Covariation (i.e., correlation).
3. Causal direction (e.g., temporal precedence; X happens before Y).
4. Non-spuriousness (um...haha...what?).

Rules for Establishing Causality

1. Variation in the causal variable
2. Covariation (i.e., correlation).
3. Causal direction (e.g., temporal precedence; X happens before Y).
4. Non-spuriousness (i.e., not due to variation in a third variable, Z, that might be causing both X *and* Y).

Necessity/Sufficiency

Necessary condition:

- Y is true only if X is true.
- X has to happen in order for Y to happen, but it's not enough to ensure Y happens.
- Showing up to class is a necessary condition for getting an A (but you won't get an A just for showing up).

Necessary condition: Sufficient condition:

- if X is true, then Y is true.
- X happening is enough for Y to happen, but Y can happen without X.
- Getting 100 percent is a sufficient condition for getting an A on the test, but you would also get an A with a 99, 98, ... , 93.

Necessary and sufficient condition:

- the rarest of all the causation Pokémon.
- X has to happen for Y to happen AND X is enough to cause Y.
- Getting every answer correct is necessary AND sufficient for getting a *perfect* score on the test.

Let's Practice

Being $\geq 35 \rightarrow$ Being President of the U.S.

Let's Practice

Being $\geq 35 \rightarrow$ Being President of the U.S. (N)

Having tear ducts $\rightarrow$ Crying about Mufasa

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Eating ghost peppers $\rightarrow$ Having a bad time

Let's Practice

Being $\geq 35 \rightarrow$ Being President of the U.S. (N)

Having tear ducts $\rightarrow$ Crying about Mufasa (N)

Eating ghost peppers $\rightarrow$ Having a bad time (S)

Being a dad $\rightarrow$ Making a dad joke

Let's Practice

Being $\geq 35 \rightarrow$ Being President of the U.S. (N)

Having tear ducts $\rightarrow$ Crying about Mufasa (N)

Eating ghost peppers $\rightarrow$ Having a bad time (S)

Being a dad $\rightarrow$ Making a dad joke (Neither)

Having a Shih Tzu $\rightarrow$ Having a dog

Let's Practice

Being $\geq 35 \rightarrow$ Being President of the U.S. (N)

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Eating ghost peppers $\rightarrow$ Having a bad time (S)

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Being a nerd $\rightarrow$ Being a professor

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Going to Sal's $\rightarrow$ Eating Gary's

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Going to Sal's $\rightarrow$ Eating Gary's (N *and* S?)